

第7章 导行电磁波

导行电磁波： 沿着导波系统传输的电磁波

导波系统： 用于引导电磁波的系统

常见的导波系统：

双绞线

平行双导线

同轴电缆

金属管波导

矩形波导

圆形波导

椭圆波导

加脊波导

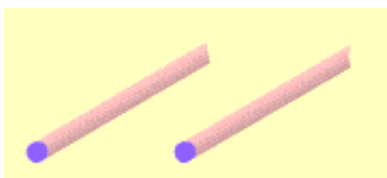
介质波导

光导纤维(光纤)

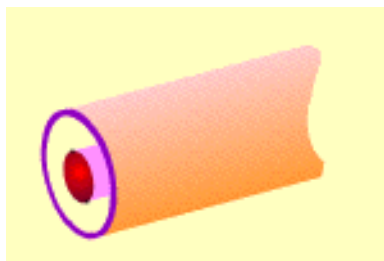
平面波导

微带线

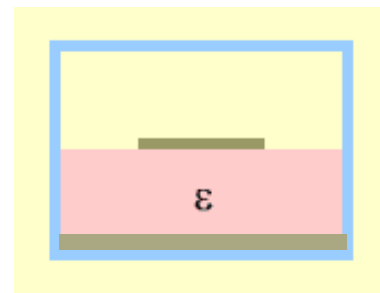
带状线



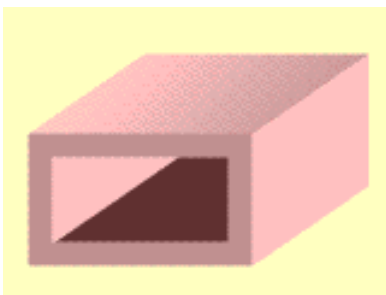
平行双导线



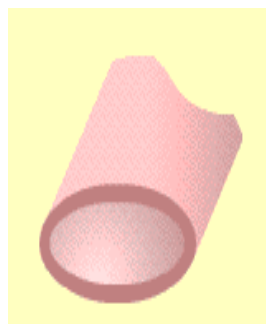
同轴电缆



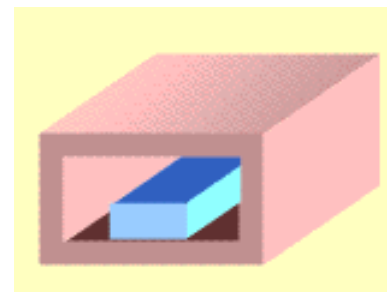
微带线



矩形波导



圆形波导



加脊波导

双导线：只用于低频波，高频时有电磁辐射 $<100\text{MHz}$ 。

同轴线：无电磁辐射，工作频带宽 $<3000\text{MHz}$ ；

微带线：用于印刷电路：分米、厘米波段。

金属波导：电磁损耗小，无辐射，主要用于厘米波、毫米波段。

介质波导：传输毫米波到光波波段。

7.1 导波系统中的TEM波与非TEM波

1) 导波系统中的场

我们所讨论的导波系统都认为是沿导波方向是均匀的
即横截面的形状都是相同的

以后为了使分析一致, 我们总是将导波系统沿 z 轴放置, 使波沿 z 方向传输

电磁场可以表示为:

$$\vec{E}(\vec{r}) = \vec{E}_0(x, y)e^{-jk_z z}$$

$$\vec{E}_0(x, y) = \hat{x}E_{x0}(x, y) + \hat{y}E_{y0}(x, y) + \hat{z}E_{z0}(x, y)$$

$$\vec{H}(\vec{r}) = \vec{H}_0(x, y)e^{-jk_z z}$$

$$\vec{H}_0(x, y) = \hat{x}H_{x0}(x, y) + \hat{y}H_{y0}(x, y) + \hat{z}H_{z0}(x, y)$$

2) 场方程

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0 \qquad \frac{\partial^2 \bar{E}}{\partial x^2} + \frac{\partial^2 \bar{E}}{\partial y^2} + \frac{\partial^2 \bar{E}}{\partial z^2} + k^2 \bar{E} = 0$$

$$\nabla^2 \vec{H} + k^2 \vec{H} = 0 \qquad \frac{\partial^2 \bar{H}}{\partial x^2} + \frac{\partial^2 \bar{H}}{\partial y^2} + \frac{\partial^2 \bar{H}}{\partial z^2} + k^2 \bar{H} = 0$$

$$\vec{E}(x, y, z) = \vec{E}_0(x, y) e^{-jk_z z}$$

$$\frac{\partial^2 \bar{E}_0}{\partial x^2} e^{-jk_z z} + \frac{\partial^2 \bar{E}_0}{\partial y^2} e^{-jk_z z} - k_z^2 \bar{E}_0 e^{-jk_z z} + k^2 \bar{E}_0 e^{-jk_z z} = 0$$

$$\nabla_t^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

$$k_c^2 = k^2 - k_z^2$$

$$\nabla_t^2 E_{z0} + k_c^2 E_{z0} = 0$$

$$\nabla_t^2 \vec{E}_0 + k_c^2 \vec{E}_0 = 0$$

$$\nabla_t^2 \vec{H}_0 + k_c^2 \vec{H}_0 = 0$$

6个标量方程

3) 导行波场分量之间的关系

$$\nabla \times \vec{H} = j\omega\epsilon\vec{E}$$

6个标量方程

$$\vec{E}(\vec{r}) = \vec{E}_0(x, y)e^{-jk_z z}$$

$$\nabla \times \vec{E} = -j\omega\mu\vec{H}$$

$$\vec{H}(\vec{r}) = \vec{H}_0(x, y)e^{-jk_z z}$$

$$\frac{\partial H_z}{\partial y} + jk_z H_y = j\omega\epsilon E_x$$

$$\frac{\partial E_z}{\partial y} + jk_z E_y = -j\omega\mu H_x$$

$$-jk_z H_x - \frac{\partial H_z}{\partial x} = j\omega\epsilon E_y$$

$$-jk_z E_x - \frac{\partial E_z}{\partial x} = -j\omega\mu H_y$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\epsilon E_z$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z$$

$$E_x = \frac{1}{k_c^2} (-jk_z \frac{\partial E_z}{\partial x} - j\omega\mu \frac{\partial H_z}{\partial y})$$

$$H_x = \frac{1}{k_c^2} (j\omega\varepsilon \frac{\partial E_z}{\partial y} - jk_z \frac{\partial H_z}{\partial x})$$

$$E_y = \frac{1}{k_c^2} (-jk_z \frac{\partial E_z}{\partial y} + j\omega\mu \frac{\partial H_z}{\partial x})$$

$$H_y = \frac{1}{k_c^2} (-j\omega\varepsilon \frac{\partial E_z}{\partial x} - jk_z \frac{\partial H_z}{\partial y})$$

$$E_{z0}(x, y), H_{z0}(x, y) \rightarrow E_{x0}, E_{y0}, H_{x0}, H_{y0}$$

$$\nabla_t^2 E_{z0} + k_c^2 E_{z0} = 0$$

纵向场法

$$TE \quad E_z = 0 \quad H_z$$

$$\nabla_t^2 H_{z0} + k_c^2 H_{z0} = 0$$

$$TM \quad H_z = 0 \quad E_z$$

4) TEM波

$$E_z = 0 \quad H_z = 0$$

$$k_c^2 = 0$$

$$k^2 - k_z^2 = 0$$

$$\nabla_t^2 \vec{E}_0 + k_c^2 \vec{E}_0 = 0$$

$$\nabla_t^2 \vec{E}_0 = 0$$

$$k_z = k = \omega\sqrt{\mu\varepsilon}$$

$$\nabla_t^2 \vec{H}_0 + k_c^2 \vec{H}_0 = 0$$

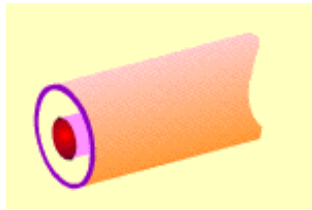
$$\nabla_t^2 \vec{H}_0 = 0$$

$$\vec{E} = \vec{E}_0(x, y)e^{-jkz}$$

$$\nabla^2 \vec{E}_0 = 0, \quad \nabla_t^2 \vec{E}_0 = 0$$

↑ $\omega \rightarrow 0$ **静电场**

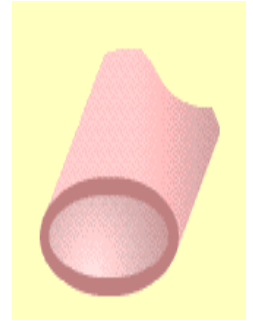
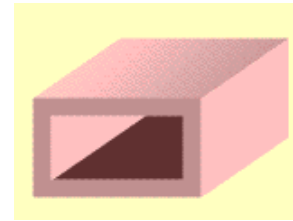
$$\nabla^2 \vec{E} + k^2 \vec{E} = 0$$



$$\nabla^2 \vec{H}_0 = 0, \quad \nabla_t^2 \vec{H}_0 = 0$$

↑ $\omega \rightarrow 0$ **恒定磁场**

$$\nabla^2 \vec{H} + k^2 \vec{H} = 0$$



$$\vec{E}_0 = 0, \quad \vec{H}_0 = 0$$

不存在 TEM波

$$\vec{E}_0 = \frac{V_0}{\ln \frac{b}{a}} \frac{\hat{\rho}}{\rho}$$

$$\vec{H}_0 = \frac{I_0}{2\pi\rho} \hat{\phi}$$

$$\int_a^b \vec{E}_0(x, y) \cdot d\vec{l} = V_0$$

$$\oint_l \vec{H}_0(x, y) \cdot d\vec{l} = I_0$$

$$V(z) = \int_a^b \vec{E}_0(x, y) e^{-jkz} \cdot d\vec{l} = V_0 e^{-jkz}$$

$$I(z) = \oint_l \vec{H}(\vec{r}) \cdot d\vec{l} = I_0 e^{-jkz}$$

7.2 TEM波传输线

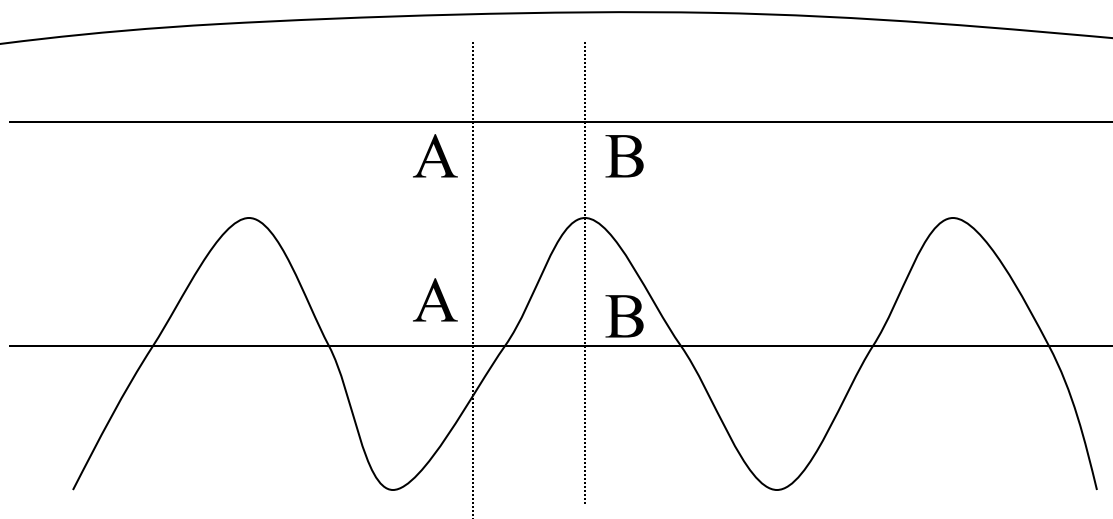
TEM 波导波系统---传输线

一、长线和短线

电长度：传输线的几何长度与其工作波长之比叫**电长度**。 l / λ

市电波长：6000km， $\frac{l}{\lambda} \ll 1$ 短线
市电输电线为短线。

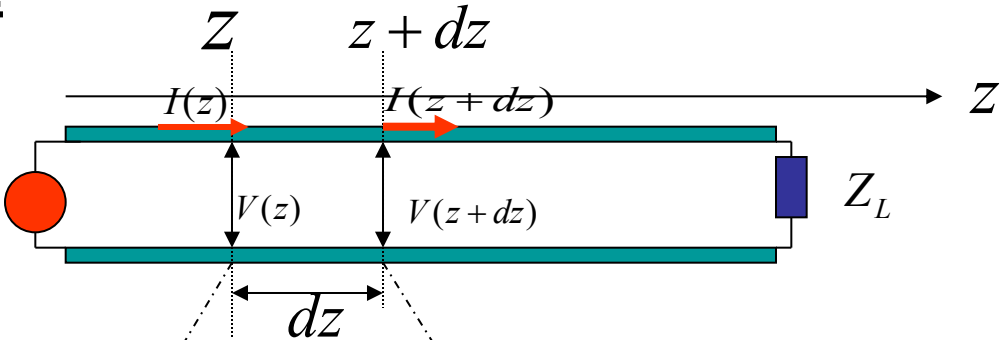
同轴电缆，一般传输米波，为长线。 $\frac{l}{\lambda} > 1$ 长线



- 低频电路中，传输线为短线，导体的电阻和电感、导体间的漏电导和电容可用**集总参数**等效。
- 高频电路中，由于传输线为长线，传输线的电阻和电感、线间的漏电导和电容不再能用集总参数等效。为**分布参数**。

2. 电报方程----长线的电路微分方程

一对导线形成的简单的电路



单位长度的

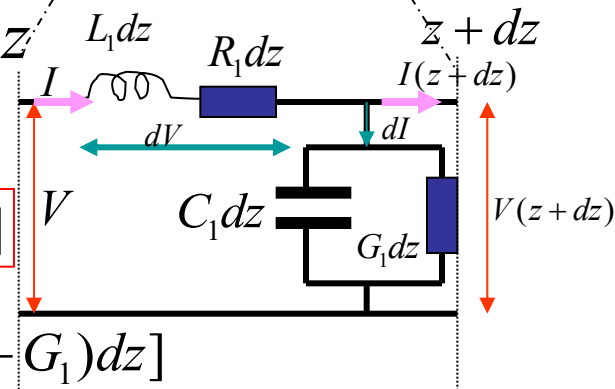
电感 L_1

电容 C_1

电阻 R_1

电导 G_1

分布参数电路模型



$$V(z) = V(z + dz) + I(z)[(j\omega L_1 + R_1)dz]$$

$$I(z) = I(z + dz) + [V(z) - dV][(j\omega C_1 + G_1)dz]$$

$$I(z) = I(z + dz) + V(z)[(j\omega C_1 + G_1)dz]$$

$Z_1 = j\omega L_1 + R_1$ 单位长度串联阻抗

$Y_1 = j\omega C_1 + G_1$ 单位长度并联导纳

$$\frac{V(z + dz) - V(z)}{dz} = \frac{dV}{dz} = -Z_1 I$$

$$\frac{I(z + dz) - I(z)}{dz} = \frac{dI}{dz} = -Y_1 V$$

电报方程(复数形式)

3. 电报方程的解

$$Z_1 = j\omega L_1 + R_1 \quad Y_1 = j\omega C_1 + G_1$$

$$\frac{dV}{dz} = -Z_1 I \quad \frac{d^2 V}{dz^2} = -Z_1 \frac{dI}{dz} = Y_1 Z_1 V \quad \frac{d^2 V}{dz^2} = \gamma^2 V \quad \gamma^2 = Y_1 Z_1$$

$$\frac{dI}{dz} = -Y_1 V \quad \frac{d^2 I}{dz^2} = -Y_1 \frac{dV}{dz} = Z_1 Y_1 I \quad \frac{d^2 I}{dz^2} = \gamma^2 I$$

$$V(z) = \underline{V_0^+} e^{-\gamma z} + V_0^- e^{\gamma z} \quad \gamma = \alpha + j\beta$$

$$\alpha = \sqrt{\frac{1}{2}[(R_1 G_1 - \omega^2 L_1 C_1) + \sqrt{(R_1^2 + \omega^2 L_1^2)(G_1^2 + \omega^2 C_1^2)}]}$$

$$\beta = \sqrt{\frac{1}{2}[-(R_1 G_1 - \omega^2 L_1 C_1) + \sqrt{(R_1^2 + \omega^2 L_1^2)(G_1^2 + \omega^2 C_1^2)}]}$$

$$V^+(z) = V_0^+ e^{-\alpha z} e^{-j\beta z}$$

γ 传播常数

α 衰减常数

β 相位常数

电压与电流的关系

$$I(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

$$\frac{dV}{dz} = -Z_1 I \qquad \frac{d(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z})}{dz} = -Z_1 (I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})$$

$$-\gamma V_0^+ e^{-\gamma z} + \gamma V_0^- e^{\gamma z} = -Z_1 I_0^+ e^{-\gamma z} - Z_1 I_0^- e^{\gamma z}$$

$$-\gamma V_0^+ = -Z_1 I_0^+ \qquad \gamma V_0^- = -Z_1 I_0^-$$

$$V_0^+ = \frac{Z_1}{\gamma} I_0^+ = Z_c I_0^+ \qquad V_0^- = -\frac{Z_1}{\gamma} I_0^- = -Z_c I_0^-$$

特性阻抗: $Z_c = \frac{V^+}{I^+} = \frac{Z_1}{\gamma} = \sqrt{\frac{Z_1}{Y_1}}$

$$I(z) = \frac{V_0^+}{Z_c} e^{-\gamma z} - \frac{V_0^-}{Z_c} e^{\gamma z}$$

4. 无损耗传输线

$$R_1 = 0 \quad G_1 = 0$$

$$\alpha = \sqrt{\frac{1}{2}[(R_1 G_1 - \omega^2 L_1 C_1) + \sqrt{(R_1^2 + \omega^2 L_1^2)(G_1^2 + \omega^2 C_1^2)}]}$$

$$\alpha = 0$$

$$\Rightarrow \beta = \omega \sqrt{L_1 C_1}$$

$$\beta = \sqrt{\frac{1}{2}[-(R_1 G_1 - \omega^2 L_1 C_1) + \sqrt{(R_1^2 + \omega^2 L_1^2)(G_1^2 + \omega^2 C_1^2)}]}$$

$$Z_c = \sqrt{\frac{Z_1}{Y_1}} = \sqrt{\frac{L_1}{C_1}}$$

$$V^+(z) = V_0^+ e^{-j\beta z}$$

$$I^+(z) = \frac{V_0^+}{Z_c} e^{-j\beta z} \quad v_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{L_1 C_1}}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{v_p}{f}$$

5. 低损耗线 $R_1 \ll \omega L_1, G_1 \ll \omega C_1$

$$\alpha \approx \frac{1}{2} \left(\frac{R_1}{Z} + G_1 Z \right) \quad \beta \approx \omega \sqrt{L_1 C_1}$$

$$Z_c \approx \sqrt{\frac{L_1}{C_1}}$$

$$v_p \approx \frac{1}{\sqrt{L_1 C_1}}$$

7.3 无耗传输线的工作状态

$$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}$$

$$I(z) = \frac{V_0^+}{Z} e^{-j\beta z} - \frac{V_0^-}{Z} e^{j\beta z} \quad V_0^- = ?$$

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{V^-(z=0)}{V^+(z=0)}$$

终端电压反射系数

在终端 $z=0$

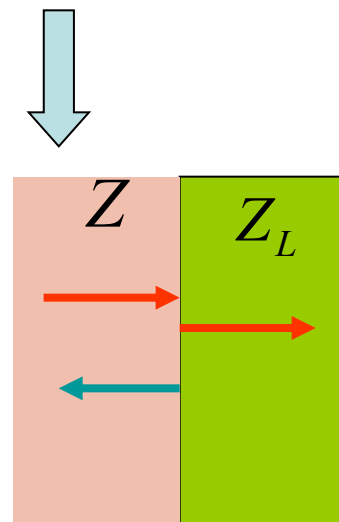
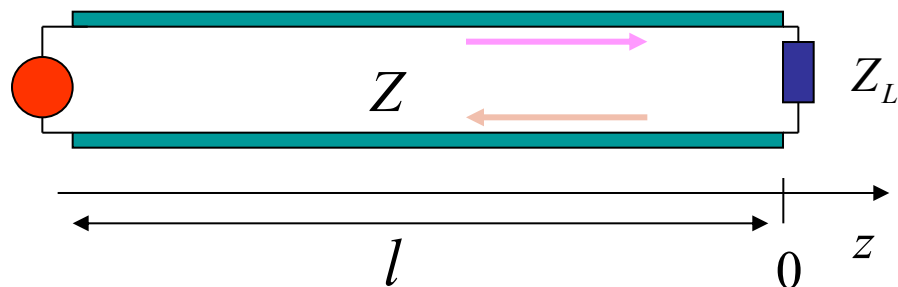
$$\frac{V(z=0)}{I(z=0)} = Z_L \Rightarrow \frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} Z = Z_L$$

$$\Rightarrow \frac{1+\Gamma}{1-\Gamma} Z = Z_L \quad \Gamma = \frac{Z_L - Z}{Z_L + Z} = |\Gamma| e^{j\theta}$$

$$V(z) = V_0^+ (e^{-j\beta z} + |\Gamma| e^{j(\beta z + \theta)})$$

$$I(z) = \frac{V_0^+}{Z} (e^{-j\beta z} - |\Gamma| e^{j(\beta z + \theta)})$$

$$Z_{in}(z=-l) = \frac{V(-l)}{I(-l)} = Z \frac{Z_L + jZ \tan(\beta l)}{Z + jZ_L \tan(\beta l)}$$



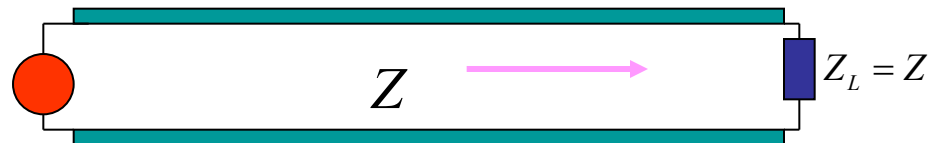
$$1) \quad Z_L = Z$$

$$\Gamma = \frac{Z_L - Z}{Z_L + Z} = 0$$

$$V(z) = V_0^+ (e^{-j\beta z} + |\Gamma| e^{j(\beta z + \theta)}) = V_0^+ e^{-j\beta z}$$

$$I(z) = \frac{V_0^+}{Z} (e^{-j\beta z} - |\Gamma| e^{j(\beta z + \theta)}) = \frac{V_0^+}{Z} e^{-j\beta z}$$

$$Z_{in}(l) = Z$$



只有入射波,无反射波
入射波全被负载电阻吸收
负载阻抗与传输线特性阻抗**匹配**

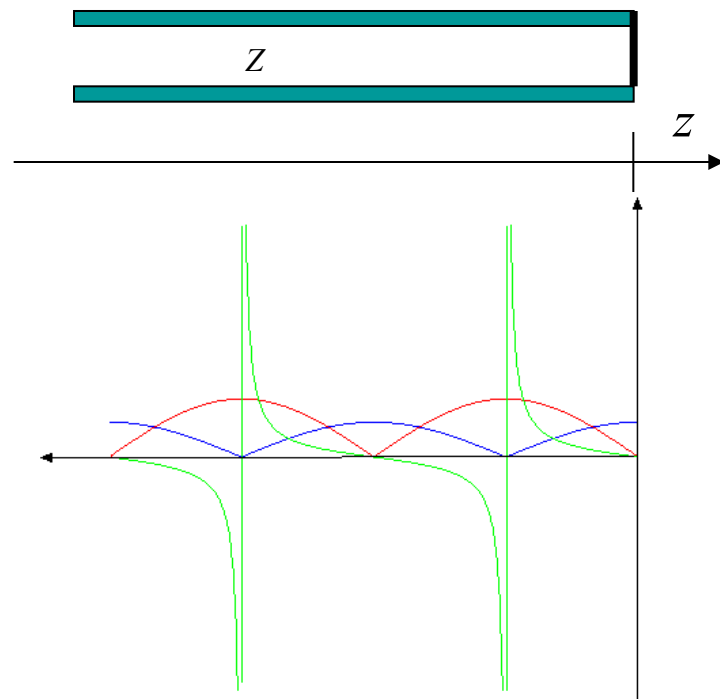
$$2) \quad Z_L = 0 \quad \text{终端短路}$$

$$\Gamma = \frac{Z_L - Z}{Z_L + Z} = -1$$

$$V(z) = -j2V_0^+ \sin(\beta z)$$

$$I(z) = 2 \frac{V_0^+}{Z} \cos(\beta z)$$

$$Z_{in}(l) = jZ \tan(\beta l)$$



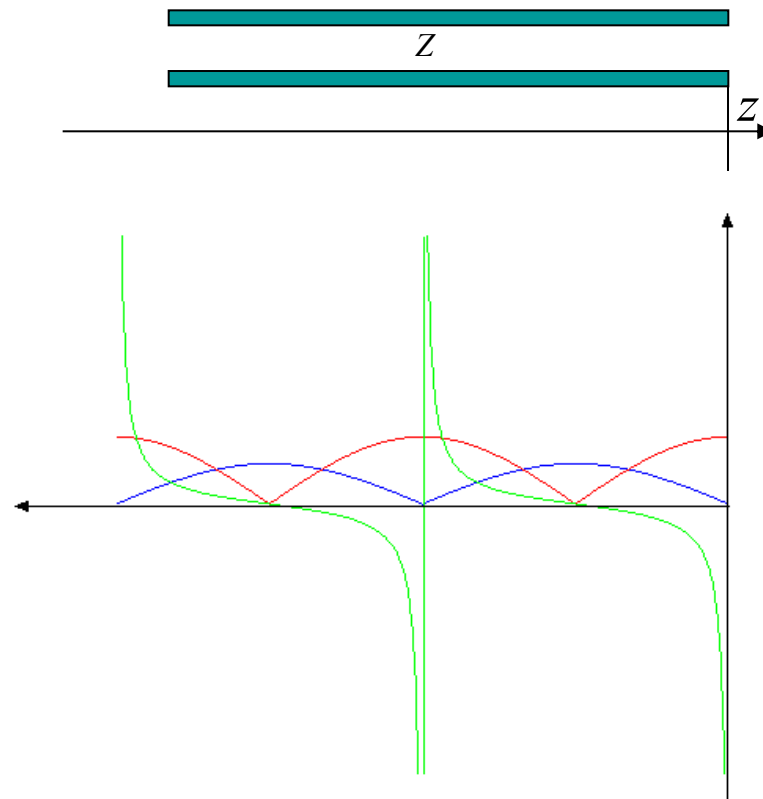
3) $Z_L = \infty$ 终端开路

$$\Gamma = \frac{Z_L - Z}{Z_L + Z} = 1$$

$$V(z) = 2V_0^+ \cos(\beta z)$$

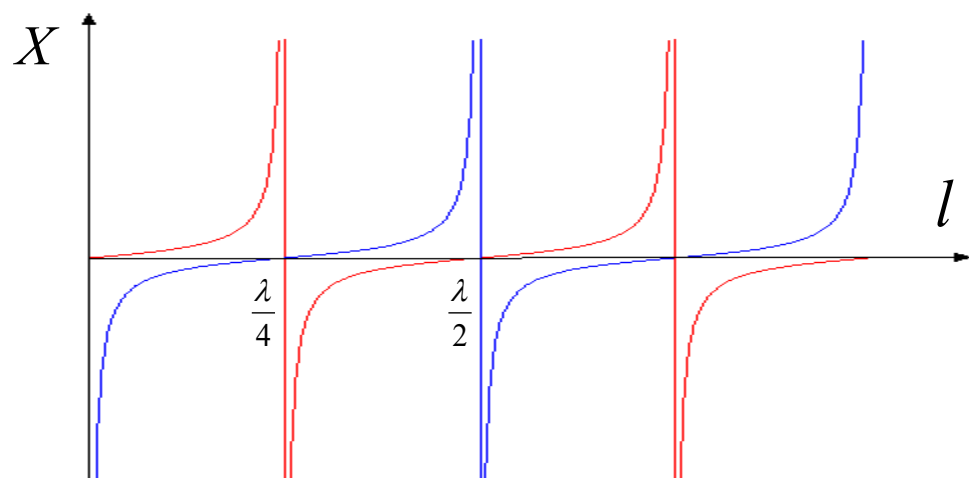
$$I(z) = -j2 \frac{V_0^+}{Z} \sin(\beta z)$$

$$Z_{in}(l) = -jZ \operatorname{ctg}(\beta l)$$

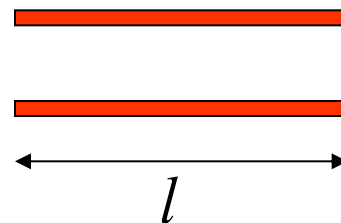


$$Z_{in}(l) = jZ_0 \tan(\beta l) \quad \text{———}$$

$$Z_{in}(l) = -jZ_0 \cot(\beta l) \quad \text{———}$$



可以用一节短路或开路线实现
 一个电容
 一个电感
 一串联谐振回路
 一并联谐振回路

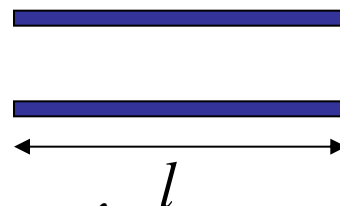


$0 < l < \frac{\lambda}{4}$ 感性负载

$\frac{\lambda}{4} < l < \frac{\lambda}{2}$ 容性负载

$l = \frac{\lambda}{4}(2n+1)$ 开路, 并联谐振

$l = \frac{\lambda}{2}n$ 短路, 串联谐振



$0 < l < \frac{\lambda}{4}$ 容性负载

$\frac{\lambda}{4} < l < \frac{\lambda}{2}$ 感性负载

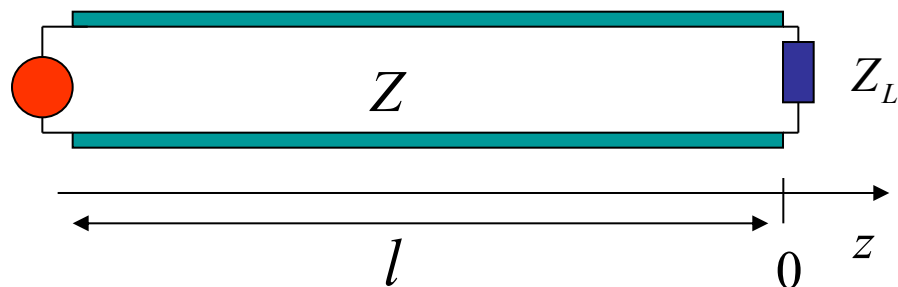
$l = \frac{\lambda}{4}(2n+1)$ 短路, 串联谐振

$l = \frac{\lambda}{2}n$ 开路, 并联谐振

4) 一般情况

$$V(z) = V_0^+ (e^{-j\beta z} + |\Gamma| e^{j(\beta z + \theta)})$$

$$I(z) = \frac{V_0^+}{Z} (e^{-j\beta z} - |\Gamma| e^{j(\beta z + \theta)})$$



驻波或驻行波

在电压波腹点:

$$2\beta l_{\max} + \theta = \pm 2n\pi$$

$$V_{\max} = V_0^+ (1 + |\Gamma|)$$

$$l_{\max} = -\frac{\theta}{4\pi} \lambda \pm n \frac{\lambda}{2}$$

$$Z_{in}(l_{\max}) = Z \frac{1 + |\Gamma|}{1 - |\Gamma|} = Z \rho$$

在电压波节点:

$$2\beta l_{\min} + \theta = \pm (2n + 1)\pi$$

$$V_{\min} = V_0^+ (1 - |\Gamma|)$$

$$l_{\min} = -\frac{\theta}{4\pi} \lambda \pm \frac{\lambda}{4} \pm n \frac{\lambda}{2}$$

$$Z_{in}(l_{\min}) = Z \frac{1 - |\Gamma|}{1 + |\Gamma|} = Z / \rho$$

电压驻波比: $\rho = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$

$$Z_{in}(z = -l) = Z \frac{1 + |\Gamma| e^{j(2\beta l + \theta)}}{1 - |\Gamma| e^{j(2\beta l + \theta)}}$$

例1: 用一段特性阻抗为50欧姆空气填充的短路线在频率为300兆赫兹时, 实现 $L=0.1\mu\text{H}$ 的电感,求短路线的长度.

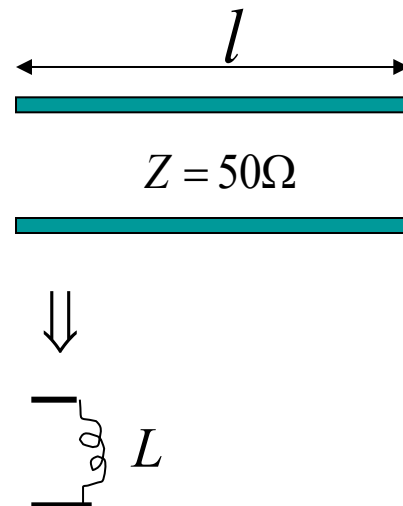
解: 短路线的输入阻抗为 $Z_{in}(l) = jZtg(\beta l)$

电感的阻抗为 $Z_{in} = j\omega L$

$$\omega L = Ztg(\beta l)$$

$$tg(\beta l) = \frac{\omega L}{Z} \quad \beta = \frac{2\pi}{\lambda}$$

$$l = \frac{\lambda}{2\pi} tg^{-1}\left(\frac{\omega L}{Z}\right) = 0.086\lambda = 0.086m$$



例2: 为特性阻抗为 50Ω 的传输线级联一节传输线, 使其与 100Ω 的负载匹配,

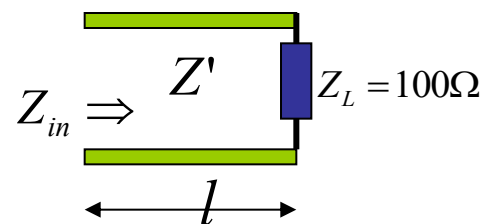
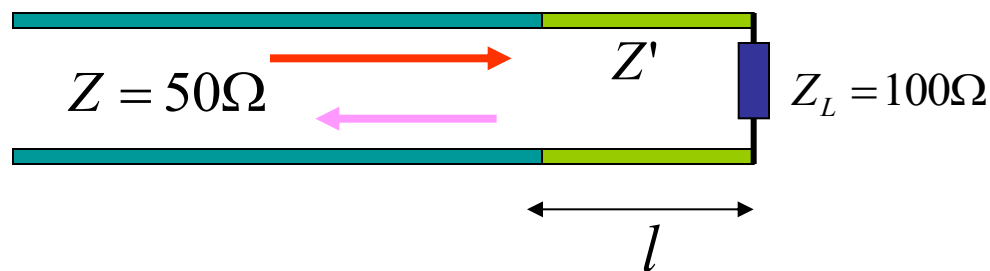
求: 级联传输线的 Z', l

解: $Z_{in} = Z$ 无反射

$$Z_{in}(l) = Z' \frac{Z_L + jZ' \operatorname{tg}(\beta l)}{Z' + jZ_L \operatorname{tg}(\beta l)}$$

$$l = \frac{\lambda}{4} \quad Z_{in} = \frac{Z'^2}{Z_L} = Z$$

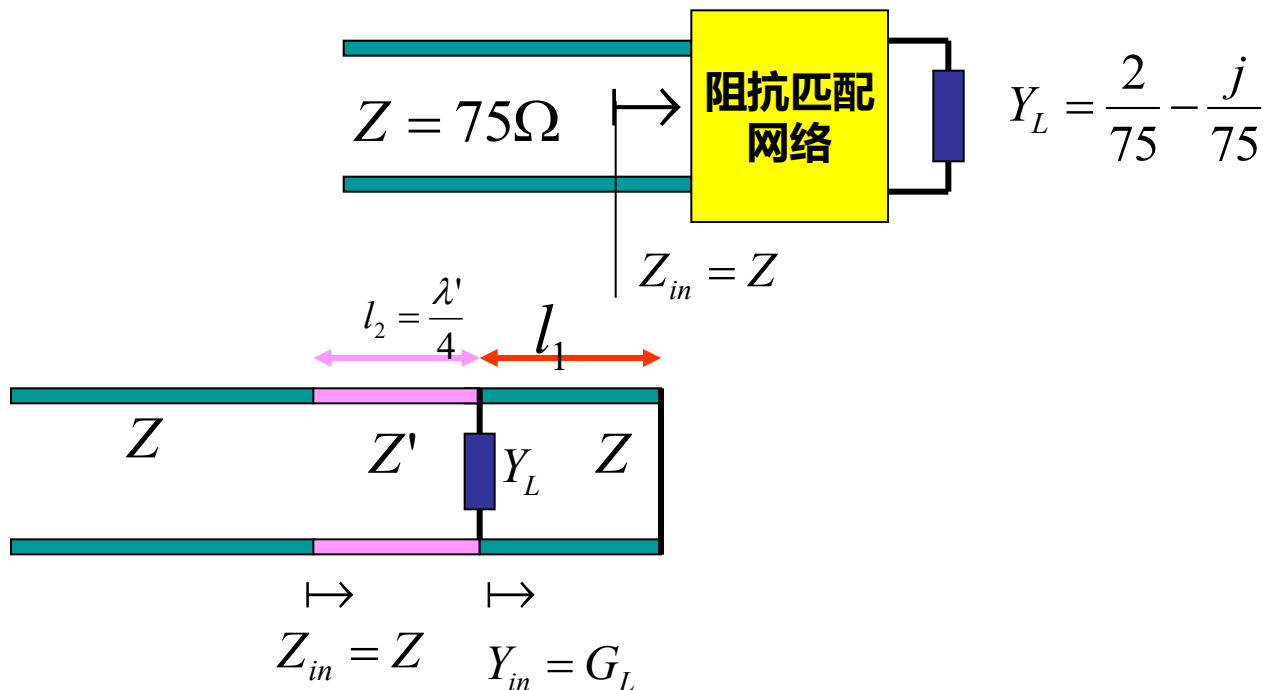
$$\begin{aligned} Z' &= \sqrt{ZZ_L} \\ &= \sqrt{50 \times 100} = 70.71\Omega \end{aligned}$$



$\frac{1}{4}$ 波长阻抗变换

例3: 为特性阻抗为 75Ω 的传输线和导纳为 $Y=(2-j)/75(S)$ 的负载进行阻抗匹配.

解:



$$l_1 = ?, Z' = ? \quad \mapsto Z'_L$$

要求短路线的输入导纳为 $Y_{in}(l_1) = \frac{j}{75}$ $Z_{in}(l_1) = -j75 = j75 \tan(\beta l_1)$

$$Z'_L = 1 / (Y_L + Y_{in}) = 75 / 2 \quad l_1 = \frac{3}{8} \lambda_1$$

$$Z' = \sqrt{Z \cdot Z'_L} = \sqrt{75 \cdot 75 / 2} = 53\Omega$$

7.4 矩形波导

矩形波导中不能传输TEM波,仅能传输非TEM波

用纵向场法,先求 E_z 和 H_z

$$TM \rightarrow E_z \quad TE \rightarrow H_z$$

1) TM波 求 E_z 在波导中电场方程为

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0 \quad k = \omega \sqrt{\mu \epsilon}$$

$$\nabla^2 E_z + k^2 E_z = 0 \quad E_z(x, y, z) = E_0(x, y) e^{-jk_z z}$$

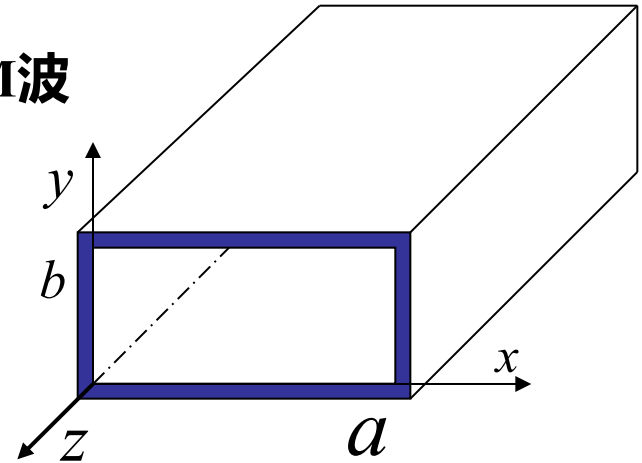
$$\nabla_t^2 E_0(x, y) + k_c^2 E_0(x, y) = 0 \quad k_c^2 = k^2 - k_z^2 \quad \nabla_t^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

用分离变量法 $E_0(x, y) = X(x)Y(y)$

$$\frac{\partial^2}{\partial x^2} E_0(x, y) + \frac{\partial^2}{\partial y^2} E_0(x, y) + k_c^2 E_0(x, y) = 0$$

$$Y(y) \frac{\partial^2 X(x)}{\partial x^2} + X(x) \frac{\partial^2 Y(y)}{\partial y^2} + k_c^2 X(x)Y(y) = 0$$

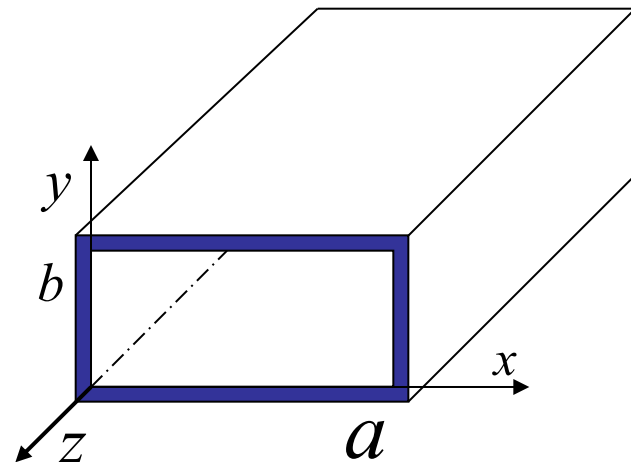
$$X''Y + XY'' + k_c^2 XY = 0$$



$$X''Y + XY'' + k_c^2 XY = 0 \quad / XY$$

$$\frac{X''}{X} + \frac{Y''}{Y} + k_c^2 = 0$$

$$\frac{X''}{X} = -k_x^2 \quad \frac{Y''}{Y} = -k_y^2 \quad k_x^2 + k_y^2 = k_c^2$$



$$X'' + k_x^2 X = 0 \quad X(x) = c_1 \cos(k_x x) + c_2 \sin(k_x x)$$

$$Y'' + k_y^2 Y = 0 \quad Y(y) = c_3 \cos(k_y y) + c_4 \sin(k_y y)$$

$$E_0(x, y) = [c_1 \cos(k_x x) + c_2 \sin(k_x x)][c_3 \cos(k_y y) + c_4 \sin(k_y y)]$$

下面利用边界条件计算积分常数

$$E_0(x, y) = [c_1 \cos(k_x x) + c_2 \sin(k_x x)][c_3 \cos(k_y y) + c_4 \sin(k_y y)]$$

近似认为波导壁是理想导体,在波导壁上 $E_t = 0$

$$E_z(x, y, z) = E_0(x, y)e^{-jk_z z} \quad E_z = 0 \rightarrow E_0 = 0$$

$$(1) x = 0 \quad E_z = 0 \rightarrow c_1 = 0$$

$$(2) y = 0 \quad E_z = 0 \rightarrow c_3 = 0$$

$$(3) x = a \quad E_z = 0 \quad \sin(k_x a) = 0 \quad k_x a = m\pi \quad k_x = \frac{m\pi}{a} \quad m = 1, 2, 3, \dots$$

$$(4) y = b \quad E_z = 0 \quad \sin(k_y b) = 0 \quad k_y b = n\pi \quad k_y = \frac{n\pi}{b} \quad n = 1, 2, 3, \dots$$

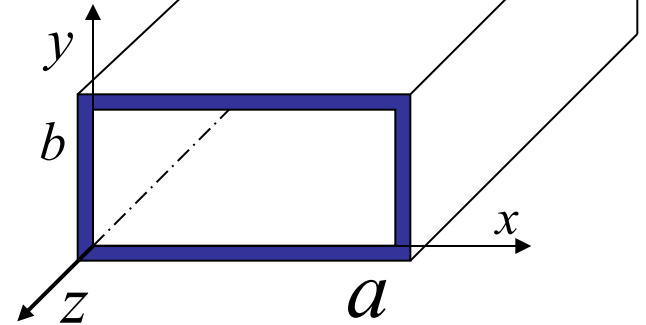
$$E_0 = c_3 c_4$$

$$E_z(x, y) = E_0 \sin\left(\frac{m\pi}{a} x\right) \sin\left(\frac{n\pi}{b} y\right) e^{-jk_z z}$$

$$k_x^2 + k_y^2 = k_c^2$$

$$k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$k_z^2 = k^2 - k_c^2$$



$$E_z = E_0 \sin\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$E_x = -j \frac{m\pi k_z}{ak_c^2} E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$E_y = -j \frac{n\pi k_z}{bk_c^2} E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_x = j \frac{\omega \epsilon n \pi}{bk_c^2} E_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_y = -j \frac{\omega \epsilon m \pi}{ak_c^2} E_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$E_x = \frac{1}{k_c^2} \left(-jk_z \frac{\partial E_z}{\partial x} - j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$E_y = \frac{1}{k_c^2} \left(-jk_z \frac{\partial E_z}{\partial y} + j\omega\mu \frac{\partial H_z}{\partial x} \right)$$

$$H_x = \frac{1}{k_c^2} \left(j\omega\epsilon \frac{\partial E_z}{\partial y} - jk_z \frac{\partial H_z}{\partial x} \right)$$

$$H_y = \frac{1}{k_c^2} \left(-j\omega\epsilon \frac{\partial E_z}{\partial x} - jk_z \frac{\partial H_z}{\partial y} \right)$$

- 波导中TM波的场为沿z方向传播的非均匀平面波；
- 不同的m、n对应不同的模式 TM_{mn} ，m、n不能为零；
- m、n分别表示在宽边和窄边上半个驻波的个数；

$$Z_{TM} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{k_z}{\omega\epsilon}$$

2) TE波

$$H_z = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_x = j \frac{m\pi k_z}{ak_c^2} H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_y = j \frac{n\pi k_z}{bk_c^2} H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$m = 0, 1, 2, \dots$$

$$E_x = j \frac{\omega\mu n\pi}{bk_c^2} H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$n = 0, 1, 2, \dots$$

$$E_y = -j \frac{\omega\mu m\pi}{ak_c^2} H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

- 波导中TE波的场为沿z方向传播的非均匀平面波；
- 不同的m、n对应不同的模式TE_{mn}，m、n不能同时为零；
- m、n分别表示在宽边和窄边上半个驻波的个数；

$$Z_{TE} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\omega\mu}{k_z}$$

$$3) k_z \quad k_z^2 = k^2 - k_c^2 \quad k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$k_z = \sqrt{k^2 - k_c^2} = k \sqrt{1 - \left(\frac{k_c}{k}\right)^2}$$

$$k > k_c \quad k_z \text{ 是实数} \quad e^{-jk_z z} \quad \text{传导模}$$

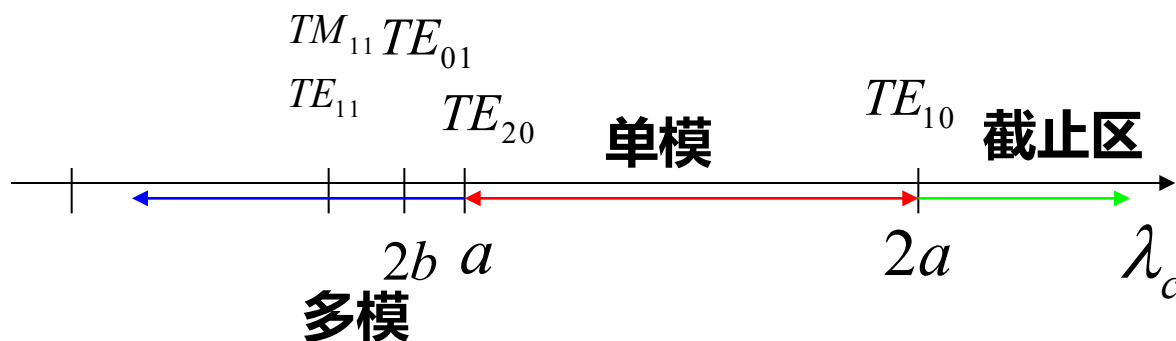
$$k < k_c \quad k_z \text{ 是虚数} \quad e^{-\alpha z} \quad \text{衰减模}$$

$$k = 2\pi f \sqrt{\mu\epsilon} \quad k > k_c \rightarrow f > \frac{k_c}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = f_c$$

模式的截止频率

$$k = \frac{2\pi}{\lambda} \quad k > k_c \rightarrow \lambda < \frac{2\pi}{k_c} = \lambda_c \quad \lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

模式的截止波长



对于传导模 $\lambda < \lambda_c$

$$v_p = \frac{\omega}{k_z} = \frac{\omega}{k \sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}} = \frac{v}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}}$$

$$\lambda_g = \frac{2\pi}{k_z} = \frac{\lambda}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}} \quad \text{波导波长}$$

$$Z_{TM} = \frac{k_z}{\omega \varepsilon} = Z \sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}$$

$$Z_{TE} = \frac{\omega \mu}{k_z} = \frac{Z}{\sqrt{1 - \left(\frac{\lambda}{\lambda_c}\right)^2}}$$

不同的模式，其相速、波导波长、波阻抗是不同的，同一模式频率不同，相速、波导波长、波阻抗也是不同的，波导为一色散系统。

例： 矩形波导 $a=22.86\text{mm}$, $b=10.16\text{mm}$
 $f=14\text{GHz}$ 的波在其中传输.

1) $\mu_r = 1, \varepsilon_r = 1$; 2) $\mu_r = 1, \varepsilon_r = 2$

波导中有多少传播模？

解： $\lambda < \lambda_c$

$$\lambda = \frac{c}{f \sqrt{\varepsilon_r}} \quad \lambda_1 = 21.43\text{mm} \quad \lambda_2 = 15.16\text{mm}$$

$$\lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

$$\lambda_{c, TE_{10}} = 2a = 45.72\text{mm} \quad \checkmark \quad \checkmark$$

$$\lambda_{c, TE_{20}} = a = 22.86\text{mm} \quad \checkmark \quad \checkmark$$

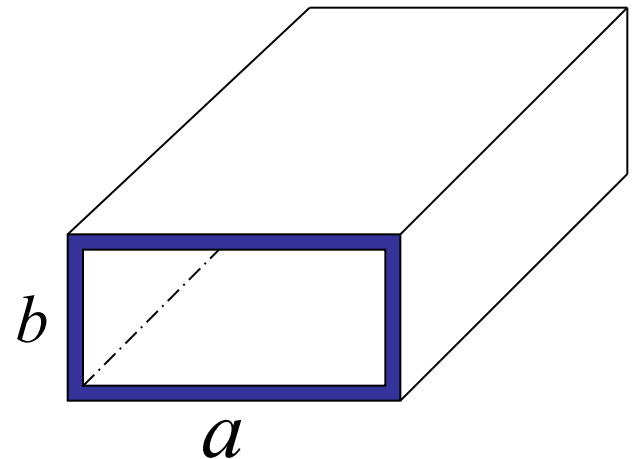
$$\lambda_{c, TE_{01}} = 2b = 20.32\text{mm} \quad \checkmark$$

$$\lambda_{c, TE_{11}, TM_{11}} = 18.56\text{mm} \quad \checkmark$$

$$\lambda_{c, TE_{30}} = 2a/3 = 15.24\text{mm} \quad \checkmark$$

$$\lambda_{c, TE_{21}, TM_{21}} = 15.19\text{mm} \quad \checkmark$$

$$\lambda_{c, TE_{02}} = b = 10.16\text{mm}$$



7.5 TE_{10} 波

实际应用时,单模传输便于向波导输送或由波导中提取能量。



单模传输时是那个模式,要满足什么条件才能仅传单模?

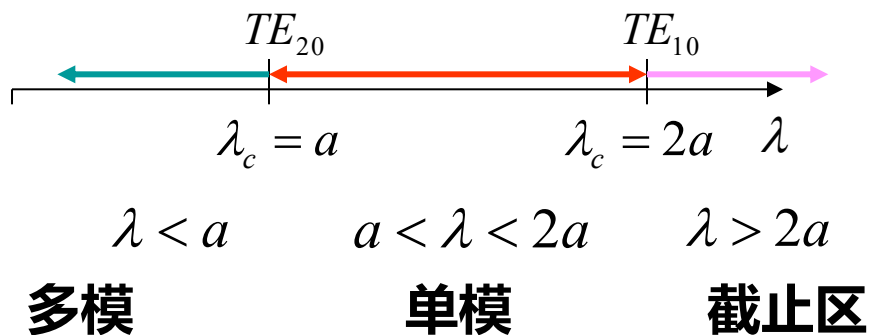
$$\lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

对于 $a > 2b$ 的矩形波导

$$TE_{10} \quad \lambda_c = 2a$$

$$TE_{20} \quad \lambda_c = a$$

其他模式: $\lambda_c < a$



认识 TE_{10} 模

场分布 $m=1, n=0$

$$H_z = H_0 \cos\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_x = j \frac{m\pi k_z}{ak_c^2} H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_y = j \frac{n\pi k_z}{bk_c^2} H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$E_x = j \frac{\omega\mu n\pi}{bk_c^2} H_0 \cos\left(\frac{m\pi}{a}x\right) \sin\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$E_y = -j \frac{\omega\mu m\pi}{ak_c^2} H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

$$H_z(x, z, t) = H_0 \cos\left(\frac{\pi}{a}x\right) \cos(\omega t - k_z z)$$

$$H_x(x, z, t) = \frac{\pi k_z}{ak_c^2} H_0 \sin\left(\frac{\pi}{a}x\right) \cos(\omega t - k_z z + \pi/2)$$

$$E_y(x, z, t) = \omega\mu \left(\frac{a}{\pi}\right) H_0 \sin\left(\frac{a}{\pi}x\right) \cos\left(\omega t - k_z z - \frac{\pi}{2}\right)$$

$$H_z = H_0 \cos\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

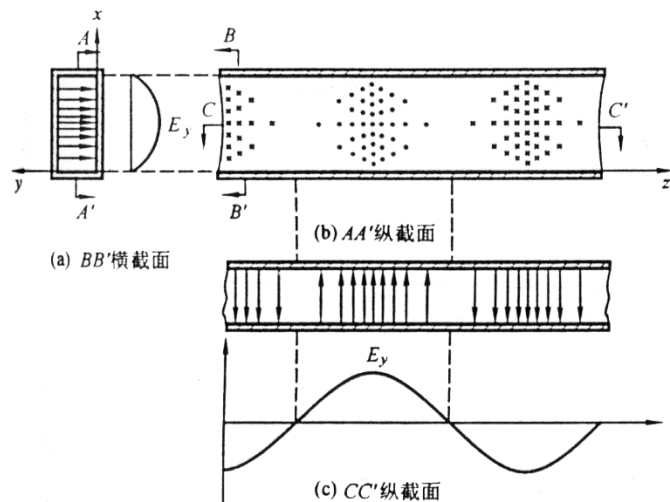
$$H_x = j \frac{\pi k_z}{ak_c^2} H_0 \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$H_y = 0$$

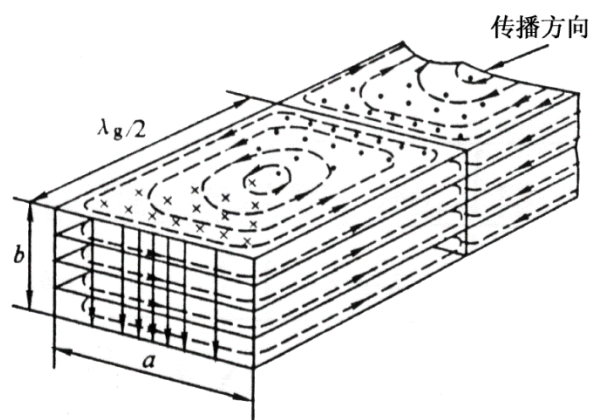
$$E_x = 0$$

$$E_y = -j \frac{\omega\mu\pi}{ak_c^2} H_0 \sin\left(\frac{\pi}{a}x\right) e^{-jk_z z}$$

$$E_y(x, z, t) = E_0 \sin\left(\frac{\pi}{a}x\right) \cos(\omega t - k_z z + \pi/2)$$



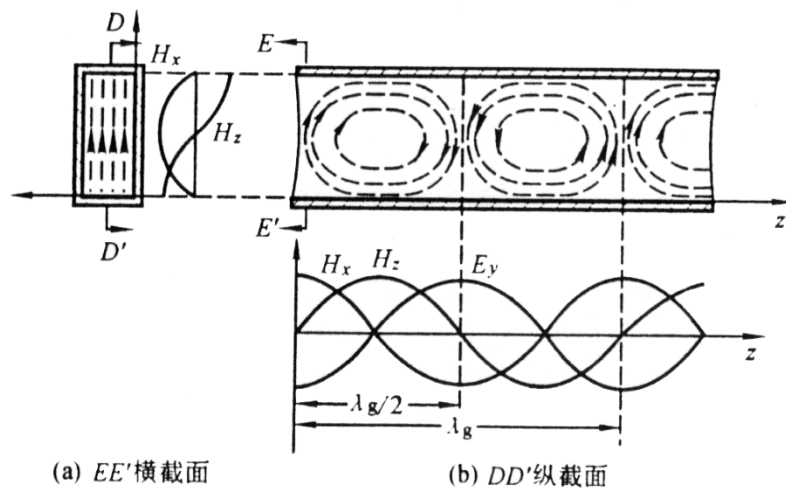
TE₁₀波的电场分布



TE₁₀波的立体电磁场分布

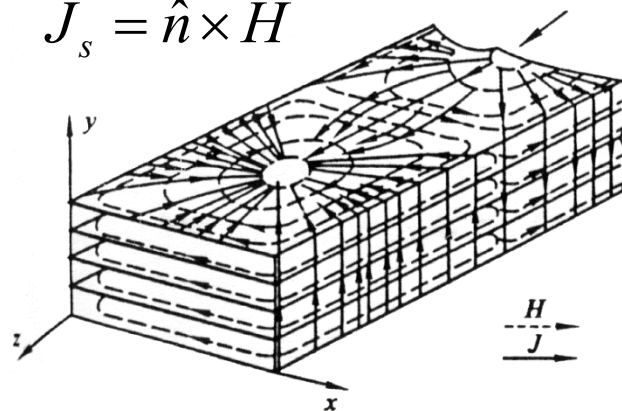
$$H_z(x, z, t) = H_0 \cos\left(\frac{\pi}{a}x\right) \cos(\omega t - k_z z)$$

$$H_x(x, z, t) = \frac{\pi k_z}{a k_c^2} H_0 \sin\left(\frac{\pi}{a}x\right) \cos(\omega t - k_z z + \pi/2)$$



TE₁₀波的磁场分布

$$\vec{J}_s = \hat{n} \times \vec{H}$$



矩形波导中TE₁₀模的管壁电流

例1:已知矩形波导尺寸,计算在什么频率范围仅传单模.

$$a \times b = 58.20\text{mm} \times 29.10\text{mm} \quad BJ-40$$

解: 单模的波长范围 $a < \lambda < 2a$ $a < \frac{c}{f} < 2a$

$$\frac{c}{2a} < f < \frac{c}{a}$$

$$2.57\text{GHz} < f < 5.15\text{GHz}$$

例2: 矩形波导 $a \times b = 22.86 \times 10.16\text{mm}^2$

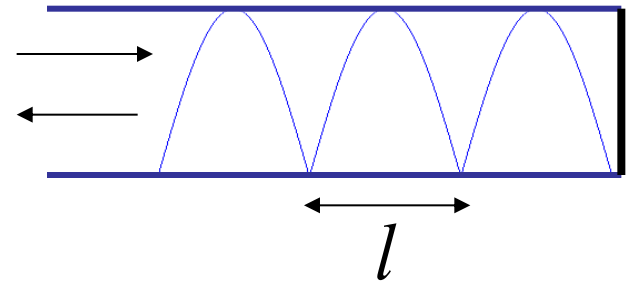
$$l = 22.5\text{mm}$$

求: f

解: $l = \frac{\lambda_g}{2} \quad \lambda_g = 2l = 45\text{mm}$

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{\lambda}{2a}\right)^2}} \quad \lambda = \frac{\lambda_g}{\sqrt{1 + \left(\frac{\lambda_g}{2a}\right)^2}} = 32\text{mm}$$

$$f = \frac{c}{\lambda} = 9.375\text{GHz}$$

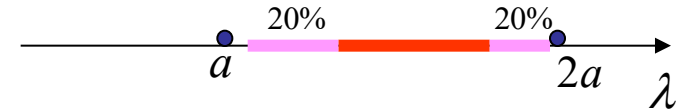


例：设计使 $f = 3 \pm 0.3\text{GHz}$ 的电磁波单模传播的波导尺寸，并使 f 与截止频率间至少还有20%的保护带。

解： $f_1 = 2.7\text{GHz}$ $f_2 = 3.3\text{GHz}$

$$2a > \lambda > a$$

增加20%的保护带，则TE10波的设计频率为：



$$f'_1 = f_1 \times (1 - 0.2) = 0.8f_1 \rightarrow \lambda_{\max} = \frac{c}{0.8f_1} < 2a$$

$$f'_2 = f_2 \times (1 + 0.2) = 1.2f_2 \rightarrow \lambda_{\min} = \frac{c}{1.2f_2} > a$$

$$a < \frac{c}{1.2f_2} = 75.75\text{mm} \quad 2a > \frac{c}{0.8f_1} = 138.9\text{mm}$$

$$a > 69.45\text{mm}$$

取 $a = 70\text{mm}, b = 10\text{mm}$

7.7 谐振腔

1. TEM传输线谐振回路

$$Z_{in1}(l_1) = jZtg(\beta l_1)$$

$$Z_{in2}(l_2) = jZtg(\beta l_2) = jZtg\left[\frac{2\pi}{\lambda}\left(n\frac{\lambda}{2} - l_1\right)\right]$$

$$= jZtg(n\pi - \beta l_1) = -jZtg(\beta l_1)$$

$$l = n\frac{\lambda}{2} = n\frac{c}{f} \frac{1}{2\sqrt{\mu_r \epsilon_r}} \Rightarrow f = n \frac{c}{2l\sqrt{\mu_r \epsilon_r}}$$

谐振腔的特点是同一尺寸的谐振腔可以在许多频率上发生谐振。

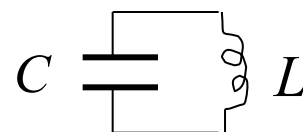
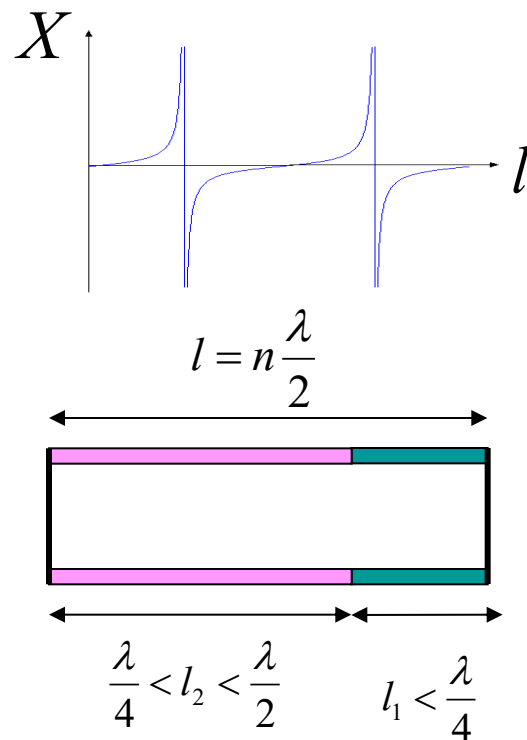
也可以说谐振腔有许多谐振模式。每一谐振模式对应一谐振频率。

传输线谐振腔中：

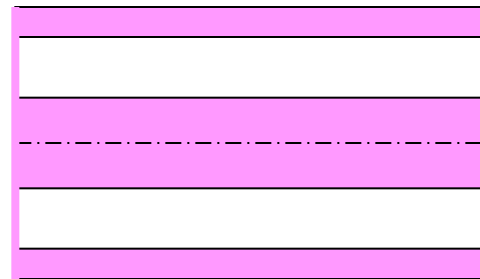
$$V(z) = -j2V_0^+ \sin(\beta z) \quad I(z) = 2\frac{V_0^+}{Z} \cos(\beta z)$$

谐振状态： $\beta l = n\pi \quad \beta = \frac{n\pi}{l}$

$$I(z) = 2\frac{V_0^+}{Z} \cos\left(\frac{n\pi}{l}z\right) \quad V(z) = -j2V_0^+ \sin\left(\frac{n\pi}{l}z\right)$$



同轴谐振腔



2. 矩形谐振腔

一段矩形波导当两端短路就形成一个矩形谐振腔。

矩形谐振腔就是一个矩形导体盒子。

矩形谐振腔中的场

设原来矩形波导中的某个场，如TE模中的 E_y 分量为

$$E_y^i = -j \frac{\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) e^{-jk_z z}$$

在 $z=0$ 短路，放理想导体平面，入射波全反射形成驻波，根据电场的边界条件，得

$$E_y = 2 \frac{\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin(k_z z)$$

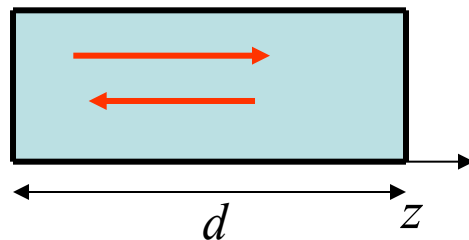
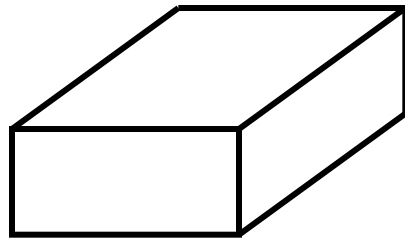
在 $z=-d$ 有理想导电壁，根据边界条件 $E_y = 0 \rightarrow \sin(k_z d) = 0 \quad k_z d = l\pi$

$$E_y = 2 \frac{\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin\left(\frac{m\pi}{a}x\right) \cos\left(\frac{n\pi}{b}y\right) \sin\left(\frac{l\pi}{d}z\right) \quad k_z = \frac{l\pi}{d} \quad l = 1, 2, 3, \dots$$

$$E_x, H_x, H_y, H_z$$

$$TE_{mnl}$$

$$TM_{mnl}$$



矩形谐振腔的谐振频率

$$k_z = \frac{l\pi}{d} \quad k_z^2 = k^2 - k_c^2 \quad k_c^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$k = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2} = 2\pi f \sqrt{\mu\varepsilon}$$

$$f = \frac{1}{2\pi\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{l\pi}{d}\right)^2}$$

一个谐振腔可谐振于很多频率!

3.品质因素Q

$$\text{固有品质因素 } Q_0 = \frac{2\pi \times \text{腔内总电磁能}}{\text{一个周期的损耗能量}} = \omega_0 \frac{W}{P_L}$$

和一般的谐振电路相比,由于谐振腔的损耗很小,因此其Q值很高.

例如: $a = b = d = 3\text{cm}$ 的铜腔,在 TE_{101} $Q_0 = 12700$

谐振腔的特点:

损耗小,Q值高;

谐振频率高, 应用于厘米波长;

制作容易。

例1: 用同轴线制作一谐振腔, 在 $f=300\text{MHz}$ 频率上谐振。求谐振腔的长度。该腔还可以在哪些频率谐振。

$$\text{解: } f = 300\text{MHz} \Rightarrow \lambda_0 = \frac{c}{f} = 1\text{m} \quad d = n \frac{\lambda}{2} = 0.5\text{m} \quad f_n = nf$$

例2: 矩形波导谐振腔内为空气, $a=b=d=3\text{cm}$, 求其谐振模式为 TE_{101} 的谐振频率。

$$f_{mnl} = \frac{1}{2\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{l}{d}\right)^2}$$

$$f_{101} = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{d}\right)^2} = 7.071\text{GHz}$$